

STOCK PREDICT

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Stock Price Forecasting with Machine Learning and Sentiment Analysis

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Chapter 1:

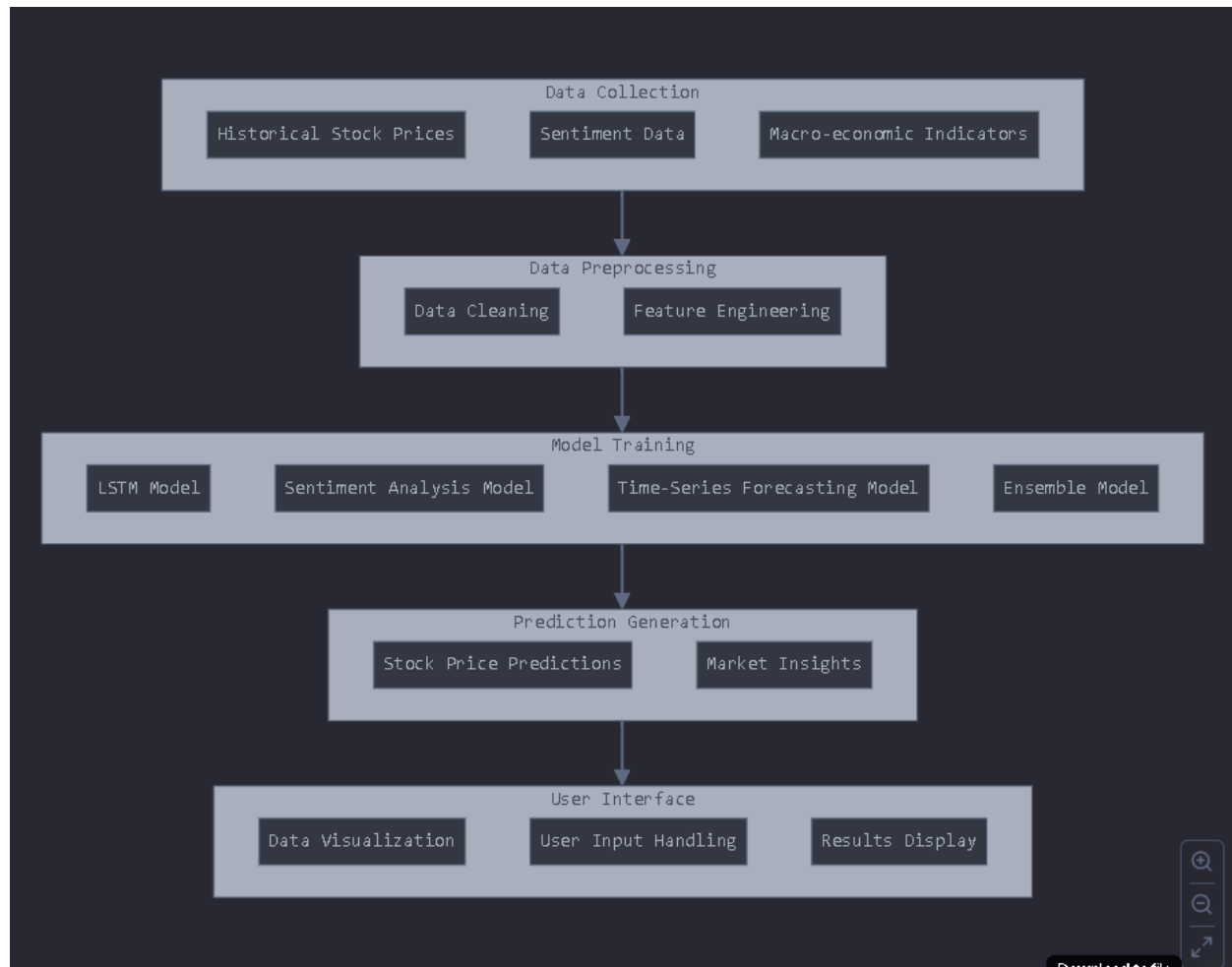
Introduction:

The purpose of this project proposal document is to outline the development of an advanced stock price forecasting system made to specifically take into account the unique characteristics of the Pakistan Stock Exchange (PSX). Conventional forecasting methods mostly base their projections on historical price trends. Therefore,, these techniques frequently fail to take into consideration abrupt changes in the market environment and the non-linear linkages that control fluctuations in stock prices. This project aims to improve upon these traditional techniques by incorporating both traditional statistical techniques and modern machine learning techniques. The primary focus is on improving stock price predictions by taking into account not only the historical stock prices but also sentiment analysis and macro-economic indicators. [5] The PSX market is influenced by a wide range of factors such as political developments, economic conditions and market sentiments. Traditional time-series forecasting methods like ARIMA often struggle to account for abrupt market changes and non-linear factors. This limitation is highlighted even more in PSX as it is a highly volatile market that is heavily influenced by external factors. The proposed system addresses these challenges by incorporating state-of-the-art machine learning techniques like recurrent neural networks (RNNs) with traditional statistical models like ARIMA or Bayesian Structural Time-Series (BST). Moreover, sentiment analysis from social media and news sources as well as macro-economic factors like Gold and Oil prices are also incorporated to increase the prediction accuracy further.

Problem Statement:

PSX is a highly unpredictable market and projections have a big impact on investing strategies. Stock price forecasting has long been a major field of research because there are so many intricately interacting variables that affect the market, such as political developments, investor mood and economic indicators which makes accurately predicting stock prices extremely difficult. Conventional forecasting methods, such ARIMA and Bayesian time series models, mostly base their projections on past price trends because of which these techniques fail to take into consideration abrupt changes in the market environment and the non-linear linkages that control fluctuations in stock

prices. This difficulty is made much more acute in the case of the Pakistan Stock Exchange (PSX) by the market's susceptibility to regional political and economic situations. The proposed system provides a solution to these difficulties by incorporating modern machine learning techniques along with traditional statistical techniques, macro-economic data and sentiment analysis to accurately predict stock prices. In recent years, machine learning has emerged as a powerful tool for handling complex data patterns, allowing for more sophisticated and adaptable forecasting models. It is feasible to extract patterns from stock market data and get deeper insights by utilizing the capabilities of neural networks, such as RNN and Long Short-Term Memory (LSTM) models. These neural network models may also help identify trends that conventional methods would overlook [1]. Furthermore, sentiment analysis may be included into stock price prediction for increased prediction accuracy. This technique analyzes the tone and sentiment of news articles, social media postings, and financial data to provide useful context.



Problem Solution:

To address the challenges outlined in the problem statement, this project proposes the development of an advanced stock price forecasting system that combines sentiment analysis and traditional time-series forecasting methods like ARIMA or Bayesian Structural Time-Series (BST) with recurrent neural networks (RNNs) in order to overcome the issues mentioned in the problem description. The model is especially made to forecast stock prices on the Pakistan Stock Exchange by taking into account historical stock prices as well as sentiment data gathered from social media sites and financial news. Additionally, macro-economic indicators like gold and oil prices are also considered to further improve the predictive accuracy. By adding outside variables like investor mood, political news, and global economic indicators, the aim is to increase the accuracy of stock price forecasts.

The project objectives are listed below:

- Develop an Advanced Stock Forecasting Model
- Incorporate Sentiment Analysis
- Leverage Macro-Economic Indicators
- Enhance Prediction Accuracy (as compared to traditional methods)
- Provide a Predictive Analysis Platform for Investors
- Fill a Market Gap
- Validate Model Performance

Scope:

Module 1- Historical Price Prediction using LSTM

Use the LSTM Recurrent Neural Network model to predict stock prices based on historical trends.

1. Train the LSTM model on historical prices.
2. Output predicted prices in a Time-Series format.

Module 2- Sentiment Analysis

Perform sentiment analysis on data gathered from different media and news sources.

1. Gather sentiment data from various media and news sources.
2. Filter into buckets i.e related to a particular stock, political data, macro-economics related etc. Identify whether it'll have an effect on global (market) or local (stock) scope.
3. Classify the data into positive/negative sentiment.
4. Use SVM or CNN with Adaboost/Catboost for optimized classification.

Module 3- Time-Series Forecasting

Perform traditional Time-Series forecasting on historical stock prices using ARIMA or Bayesian Structural Time-Series (BST).

1. Apply the ARIMA/BST for Time-Series Forecasting.
2. Output the forecasted results.

Module 4- Front-End

Develop a user-friendly front-end for the web application.

1. Use React for handling front-end related operations.
2. Add data visualization enhancements to aid users in understanding complex data such as stock prediction graphs, sentiment analysis cues etc

Module 5- Back-End

Develop a comprehensive back-end for API and database handling.

1. Use Flask and Gunicorn for API handling.
2. Use Docker for containerization.
3. Use Nginx/Caddy to implement a reverse proxy for the website.

Module 6- Ensemble Model Architecture

Merging Sentiment Analysis, LSTM and BST (or ARIMA) Outputs through a Dense Layer/CNN

1. Feature 1: Multi-input model architecture combining LSTM, Sentiment Analysis, and BST (or ARIMA) outputs.

2. Feature 2: Flexible integration using either Dense layers or CNN for advanced feature extraction

User Classes and Characteristics:

User Class	Characteristics	Technical Expertise	Use Case
Individual Investors	Active in stock market, rely on predictive tools for making informed decisions	Low to Medium	Use the platform to get stock price predictions and sentiment analysis to guide personal investments.
Financial Analysts	Professionals analyzing market trends, interested in detailed insights and prediction accuracy.	High	Utilize advanced prediction tools and sentiment analysis to prepare financial reports and forecasts.
Stock Brokerage Firms	Require high accuracy and real-time updates for stock market trends to support client investments.	High	Leverage prediction accuracy and market trends to offer advisory and trading services to clients.
Market Researchers	Conducting market studies, focus on long-term trends and macroeconomic factors	Medium	Use forecasts for research reports, market trends analysis, and long-term predictions.
Regulatory Bodies	Monitor market conditions to ensure fair trading practices, interested in large-scale market insights.	Medium	Use market forecasts and sentiment analysis for market regulation and policy-making.
General Public	Non-expert users curious about stock	Low	Access simple forecasts and market

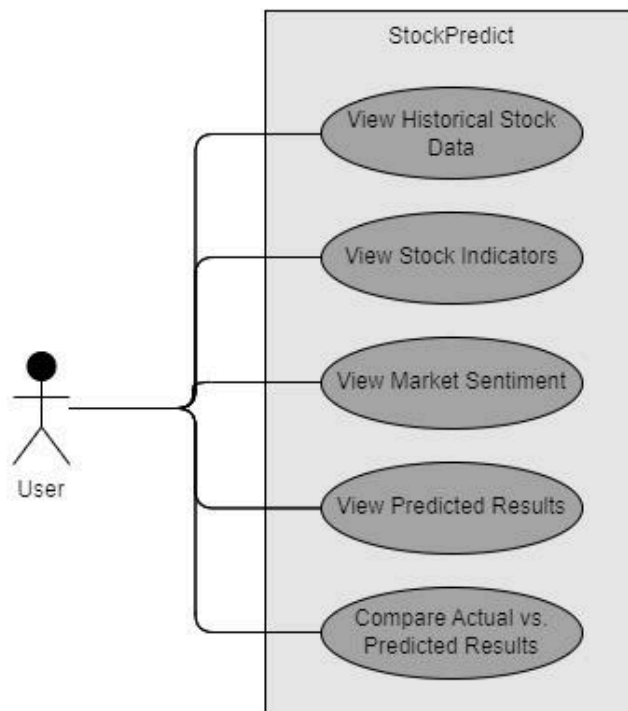
	market trends but not actively involved in trading.		sentiment to stay informed about market trends and developments.
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Chapter 2:

Project Requirement:

Use Case Diagram:

Our use case diagram will allow you to get an overview of how our project is going to entertain the users and what functionalities it will provide to its users. The integration will provide more clarity on how the user will entertain each use case.



Event Response Table:

The **Event-Response Table** is a key element in understanding the interaction between users and the system in the StockPredict application. This table outlines the different events that users may trigger while interacting with the system, the state of the system during those events, and the system's expected response. The purpose of the Event-Response Table is to ensure that all possible user interactions are well-documented and that the system reacts appropriately to each action, providing a seamless user experience. By mapping out each event and corresponding response, this table aids in identifying system functionality, performance expectations, and ensuring that all critical features are implemented effectively.

Event	System State	System Response
View Historical Stock Data	The user selects the option to view historical stock data.	System retrieves and displays historical stock price data from the database or API for the selected stocks.
Analyze Stock Indicators	The user requests analysis of specific stock indicators (e.g., moving averages, volume, etc.).	System performs calculations and displays relevant stock indicators on the UI.
View Market Sentiment	User initiates a request to view market sentiment based on news or social media.	System retrieves sentiment data, processes it through the sentiment analysis model, and displays the results.
View Predicted Results	User initiates a request to view market sentiment based on news or social media.	System runs the prediction model (e.g., LSTM or ARIMA), processes the input data, and displays the prediction.
Compare Actual vs. Predicted Results	User wants to compare actual stock prices with predicted values for a selected time period.	System retrieves actual and predicted values, plots them for comparison, and highlights any discrepancies.

Functional Requirements:

Here's the functional requirements formatted as per your request for our **StockPredict** system:

Historical Stock Data Module

- **FR-1:** The system shall retrieve historical stock data from various stock exchanges, including PSX, through an API or database.
- **FR-2:** The system shall allow users to select and filter historical data based on stock symbols, time periods, or specific events.
- **FR-3:** The system shall display the retrieved historical data in tabular and graphical formats.

Stock Indicators Analysis Module

FR-4: The system shall calculate stock indicators such as moving averages, Bollinger Bands, and RSI for selected stocks.

FR-5: The system shall provide users with a graphical analysis of stock performance based on the calculated indicators.

FR-6: The system shall allow users to customize the parameters for each indicator (e.g., periods for moving averages).

Market Sentiment Module

FR-7: The system shall extract market sentiment from news articles, social media posts, and financial reports.

FR-8: The system shall process and analyze sentiment data using different models.

FR-9: The system shall display sentiment analysis results in an easy-to-understand format, such as positive/negative sentiment scores.

Stock Price Prediction Module

FR-10: The system shall utilize machine learning models (e.g., LSTM, ARIMA) to predict future stock prices.

FR-11: The system shall allow users to input custom parameters such as forecast periods and stock symbols before making predictions.

FR-12: The system shall display predicted stock prices in graphical format, alongside historical price trends.

Actual vs. Predicted Comparison Module

FR-13: The system shall retrieve actual stock prices for a given period to compare with predicted values.

FR-14: The system shall generate comparative graphs to highlight the differences between actual and predicted stock prices.

FR-15: The system shall calculate prediction accuracy and display it alongside the comparative graph.

Non-Functional Requirements:

Here are some Non-Functional Requirements for our **StockPredict** system:

Performance

NFR-1: The system shall retrieve and display historical stock data within a few seconds of the user's request.

NFR-2: The system shall process stock predictions for selected stocks.

NFR-3: The system shall handle multiple concurrent users without significant degradation in performance.

Usability

NFR-4: The system's user interface shall be intuitive and easy to navigate, with no more than three clicks needed to access any major feature.

NFR-5: The system shall provide clear and consistent error messages to guide users in resolving issues.

NFR-6: The system shall offer responsive design, ensuring full functionality on mobile devices, tablets, and desktop browsers.

Reliability

NFR-7: The system shall maintain an uptime, ensuring availability for stock analysis and predictions.

NFR-8: The system shall save all user inputs and predictions to prevent data loss in case of system crashes or network interruptions.

NFR-9: The system shall implement backup mechanisms to restore the latest data within 2 minutes of a failure.

Scalability

NFR-10: The system shall be able to scale horizontally to accommodate increases in user traffic and data load.

NFR-11: The system shall support the addition of new stock markets and prediction models with minimal configuration changes.

Security

NFR-12: The system shall encrypt all user data, including stock preferences and prediction history, both at rest and in transit.

NFR-13: The system shall implement role-based access control (RBAC) to ensure that sensitive data and features are only accessible to authorized users.

NFR-14: The system shall automatically log out inactive users after 10 minutes of inactivity.

Maintainability

NFR-15: The system's codebase shall be modular and adhere to industry-standard coding practices to facilitate future updates and maintenance.

NFR-16: The system shall include comprehensive logging to help developers diagnose and fix issues efficiently.

Portability

NFR-17: The system shall be deployable on multiple platforms, including cloud environments (AWS, Azure) and on-premises servers.

NFR-18: The system shall support cross-browser compatibility, working seamlessly on Chrome, Firefox, Safari, and Edge.

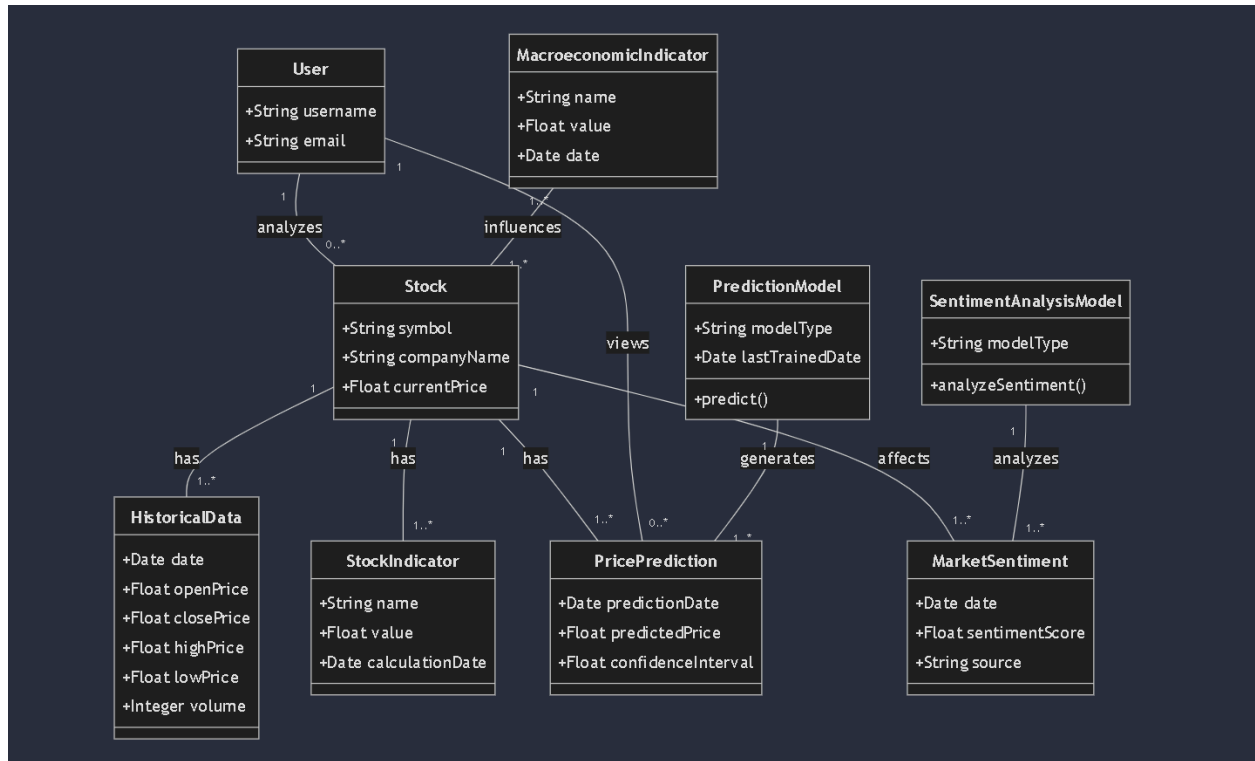
Legal & Compliance

NFR-19: The system shall comply with GDPR regulations for handling user data within the EU.

NFR-20: The system shall ensure that all financial data sources used for predictions are compliant with relevant market regulations and licensing requirements.

Domain Diagram:

In the case of the **StockPredict** system, the domain diagram outlines essential entities like stocks, indicators, market sentiment, users, and predictions. It shows how these elements connect, such as how a user interacts with historical data, predictions, and sentiment analysis, enabling the system to deliver accurate stock forecasts based on user inputs and market conditions. This diagram serves as a blueprint for understanding the core data structures and interactions required for system implementation.



Chapter 3:

System Overview:

Design:

Background:

Architectural Design:

The **StockPredict** system is designed with a modular and scalable architecture to efficiently provide stock price predictions using machine learning techniques. It is structured into distinct layers to ensure flexibility, maintainability, and high performance, suitable for handling stock data processing and prediction tasks.

System Layers Overview

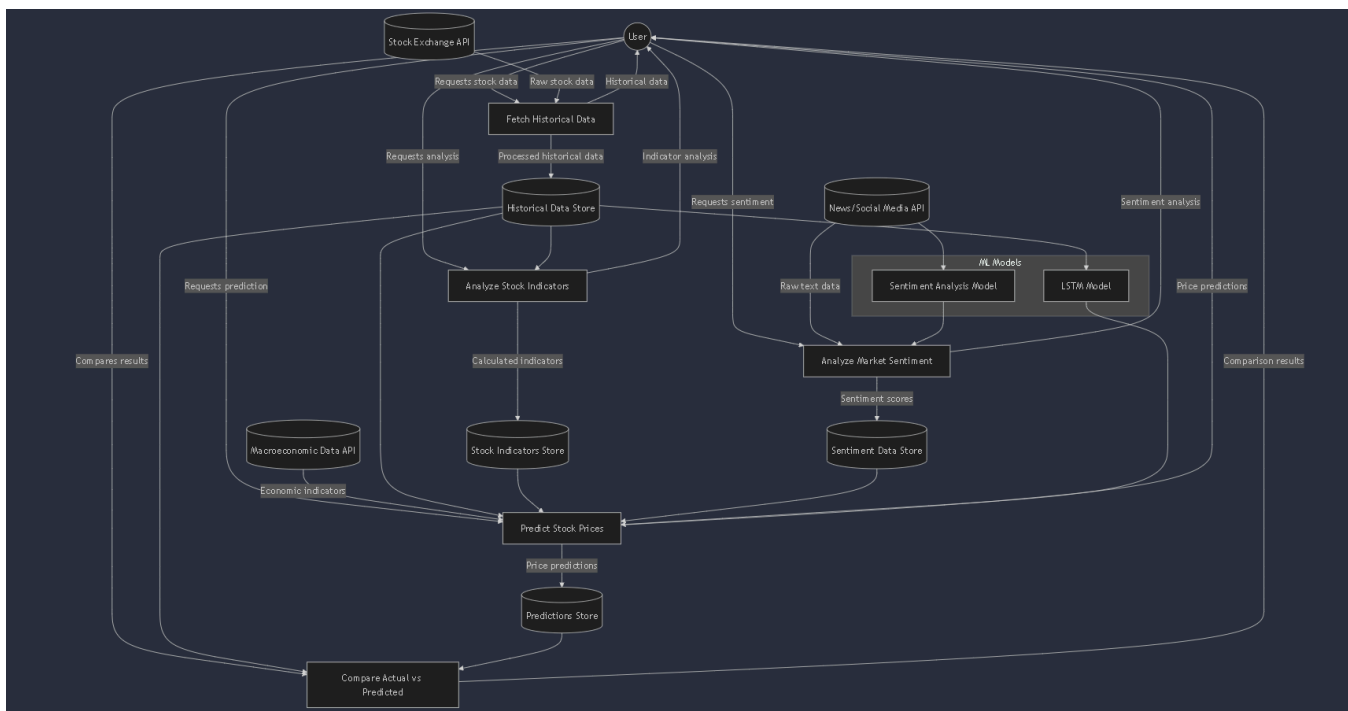
- **Presentation Layer (Front-End):**
 - The front-end is built using **React**, enabling users to interact with the system by providing stock symbols, date ranges, and viewing prediction results. The interface displays data visualizations like stock trends and comparison graphs for ease of understanding.
 - **Responsiveness:** Ensures full functionality across various devices (desktop, mobile, tablet).
- **Application Layer (Back-End):**
 - The back-end is implemented using **Flask** and **Gunicorn**, handling requests from the front-end and managing business logic.
 - **Modules:**
 - **Data Acquisition Module:** Collects historical stock data from external APIs and databases.
 - **Prediction Engine:** Uses **LSTM** (Long Short-Term Memory) neural networks to predict future stock prices based on historical data trends.
 - **Comparison Module:** Compares predicted stock prices against actual prices, providing accuracy metrics and performance insights.

- **Data Layer (Storage):**

- **Database:** A relational database such as **PostgreSQL** stores historical stock data, prediction results, and user-specific information.
- **Cloud Storage:** For larger datasets, the system can integrate with cloud storage platforms like **AWS S3** to handle scalability and data retrieval efficiently.

Design Models:

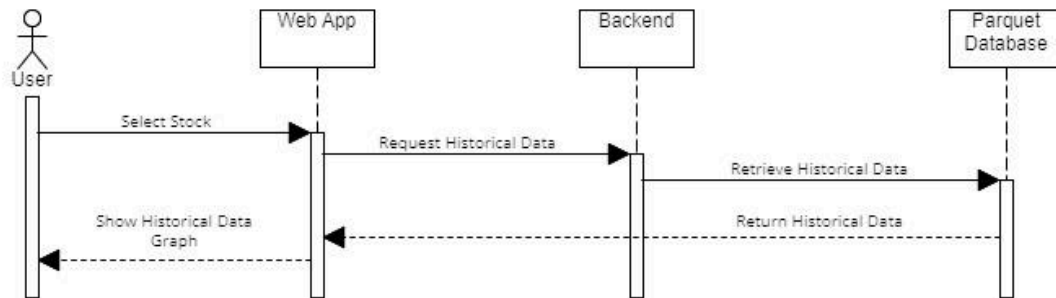
Data Flow Diagram:



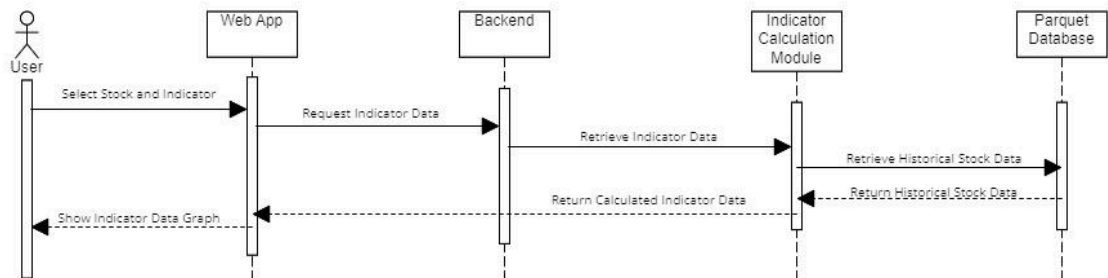
Sequence Diagrams:

Here are the sequence diagrams of all the use cases that our program is offering.

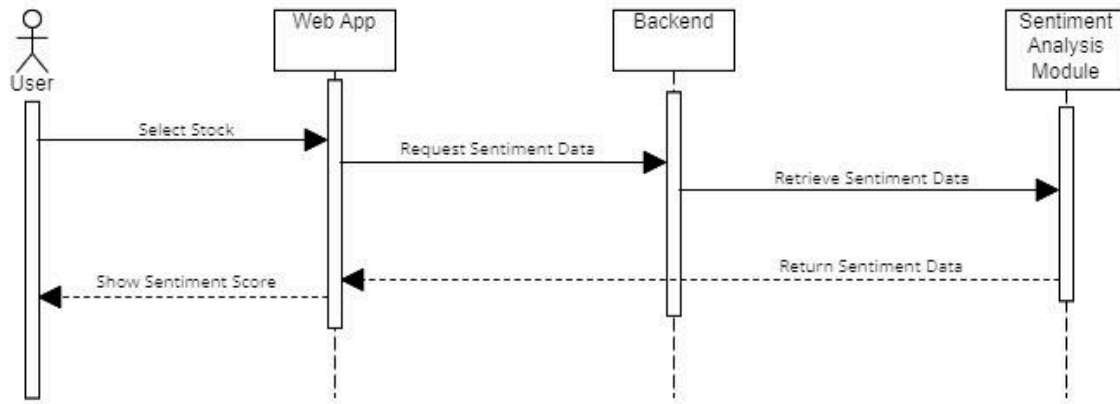
View Historical Data



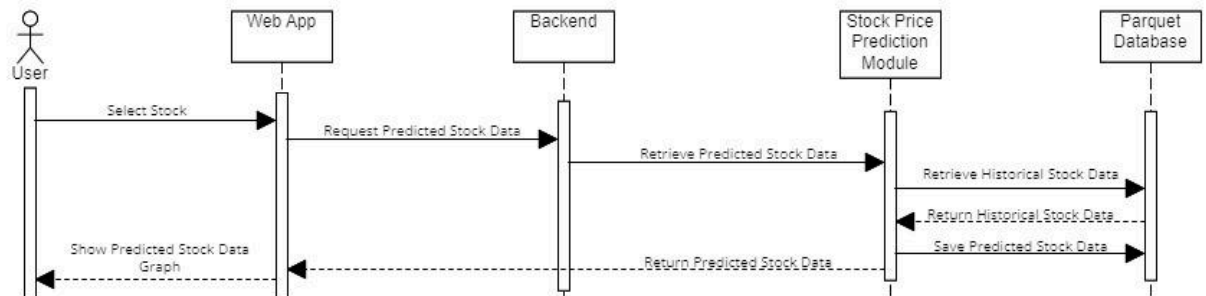
View Stocks Indicators



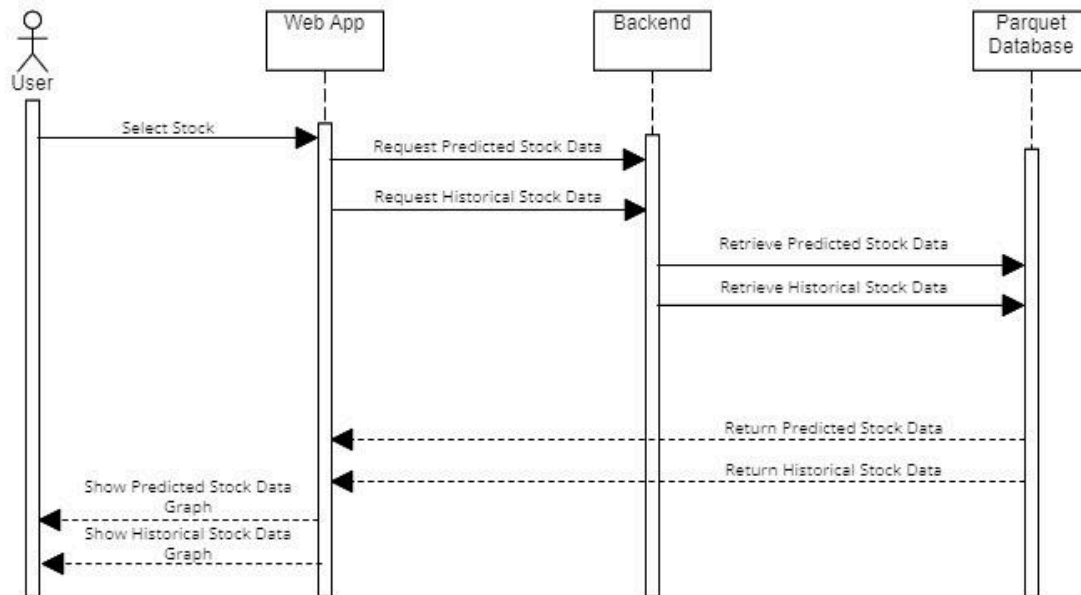
View Market Sentiments



View Predicted Results

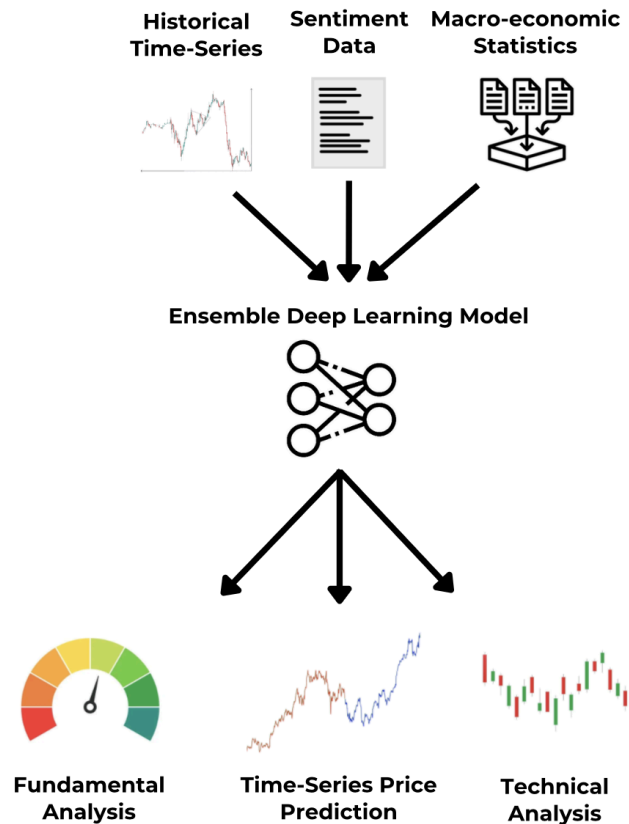


Compare Actual Results Vs Predicted Results



Architecture Diagram:

The Architecture Diagram provides a comprehensive overview of the system's structural design.



State Diagram:

A **State Diagram** for the **StockPredict** project outlines the different states the system transitions through as users interact with the stock prediction functionality. It demonstrates how the system responds to events such as requests for stock data, market analysis, and predictions. Beginning in an initial idle state, the system transitions through phases like **Loading Data**, **Performing Analysis**, and **Generating Predictions**, depending on user actions. Each state represents a specific operational mode of the system, ensuring clarity on how the system handles different processes and reacts to user inputs or external triggers

